

Multiple Choice (10 marks)

1. According to Hooke's law, if you hang by a tree branch and note how much it bends, then hanging with twice the weight produces

(a) three times the bend.
(b) twice the bend.
(c) four times the bend if the branch doesn't break.
(d) no bend at all.
(e) a bend that is unpredictable.

2. Thermal conduction involves mainly

(a) isotopes
(b) ions
(c) mesons
(d) protons
(e) photons

3. A plano-convex lens in air has a power of 5 diopters. What is the power of the lens in water? The index of refraction of the glass in the lens is 1.60.

(a) 0.83 diopters
(b) 3.3 diopters
(c) 5.0 diopters
(d) 1.0 diopters
(e) 1.2 diopters

4. How long would it take a car, starting from rest and accelerating uniformly in a straight line at 5 m/s^2 , to cover a distance of 200 m ?

(a) 18.0 s
(b) 15.5 s
(c) 9.0 s
(d) 8.0 s
(e) 10.5 s

5. A nucleus of mass M emits a γ ray of energy E_γ . The nucleus was initially at rest. What is its recoil energy?

6. $E_\gamma^2/4Mc^2$

7. $2E_{\gamma}^2/Mc^2$
8. E_{γ}/Mc^2
9. $3E_{\gamma}^2/2Mc^2$
10. $E_{\gamma}/4Mc^2$

True / False (10 marks)

Answer True or False

6. True or False: The force exerted on the block by the table is parallel to the surface of the table during the dragging motion.
7. True or False: The atomic number of the resulting element does not change after emitting 1 alpha particle and 2 beta particles.
8. True or False: The object distance for a concave mirror with the given parameters is 14.7 m, which would produce a similar image distance.
9. True or False: The specific heat capacity of sand is lower than that of vegetation, contributing to its faster cooling at night.
10. True or False: A common nuclear fission reactor heats water to generate steam for electricity production.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process of light transmission through two ideal linear polarizers oriented at a 45-degree angle to each other. Calculate and explain the percentage of intensity of unpolarized light that emerges after passing through both polarizers.
12. Discuss the motion of a body projected upward with an initial speed of 1000 cm/s. Calculate the maximum height it will reach and analyze the factors that influence this height.

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